

TUGAS PENDAHULUAN MODUL 9



Disusun Oleh :

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DIREKTORAT KAMPUS PURWOKERTO

TELKOM UNIVERSITY PURWOKERTO

2025

- Membuat project tpmodul9_2311104065 dan melakukan commit

```
C:\Users\user\source\repos\KPL_Rengganis-Tantri-Pramudita_2311104065_SISE-07-02\09_API_Design_dan_Construction_Using_Swagger\tpmodul9_2311104065>git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    deleted:    .vs/ProjectEvaluation/tpmodul9_2311104065.metadata.v9.bin
    deleted:    .vs/ProjectEvaluation/tpmodul9_2311104065.projects.v9.bin
    deleted:    .vs/ProjectEvaluation/tpmodul9_2311104065.strings.v9.bin
    deleted:    .vs/tpmodul9_2311104065/DesignTimeBuild/.dtbcache.v2
    deleted:    .vs/tpmodul9_2311104065/FileContentIndex/08a7fda9-cla8-459e-b432-4a3283f63cb6.vsidx
    deleted:    .vs/tpmodul9_2311104065/FileContentIndex/1f703fb6-fe09-46eb-b858-b535c0edaa2c.vsidx
    deleted:    .vs/tpmodul9_2311104065/FileContentIndex/306abfa1-44e0-49bb-9ca9-3776f08dd764.vsidx
    deleted:    .vs/tpmodul9_2311104065/FileContentIndex/3ed231b5-e0a2-469d-8abb-f561b215c9d0.vsidx
    deleted:    .vs/tpmodul9_2311104065/FileContentIndex/86ad0b6e-84ab-4c1f-a479-4ed48d400a50.vsidx
    deleted:    .vs/tpmodul9_2311104065/v17/.fudcache.v2
    modified:   .vs/tpmodul9_2311104065/v17/.suo
    deleted:    .vs/tpmodul9_2311104065/v17/DocumentLayout.backup.json
    modified:   .vs/tpmodul9_2311104065/v17/DocumentLayout.json
    modified:   tpmodul9_2311104065.sln
    modified:   tpmodul9_2311104065/Properties/launchSettings.json
    modified:   tpmodul9_2311104065/tpmodul9_2311104065.http

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    .vs/tpmodul9_2311104065/FileContentIndex/f02216fa-592e-47c4-90c5-e94917138660.vsidx

no changes added to commit (use "git add" and/or "git commit -a")
```

```
C:\Users\user\source\repos\KPL_Rengganis-Tantri-Pramudita_2311104065_SISE-07-02\09_API_Design_dan_Construction_Using_Swagger\tpmodul9_2311104065>git add .
error: open("09_API_Design_dan_Construction_Using_Swagger\tpmodul9_2311104065/.vs/tpmodul9_2311104065/FileContentIndex/f02216fa-592e-47c4-90c5-e94917138660.vsidx"): Permission denied
error: unable to index file '09_API_Design_dan_Construction_Using_Swagger\tpmodul9_2311104065/.vs/tpmodul9_2311104065/FileContentIndex/f02216fa-592e-47c4-90c5-e94917138660.vsidx'
fatal: adding files failed

C:\Users\user\source\repos\KPL_Rengganis-Tantri-Pramudita_2311104065_SISE-07-02\09_API_Design_dan_Construction_Using_Swagger\tpmodul9_2311104065>git add .

C:\Users\user\source\repos\KPL_Rengganis-Tantri-Pramudita_2311104065_SISE-07-02\09_API_Design_dan_Construction_Using_Swagger\tpmodul9_2311104065>git commit -m"up gitignore"
[main 485dbc3] up gitignore
15 files changed, 14 insertions(+), 14 deletions(-)
delete mode 100644 09_API_Design_dan_Construction_Using_Swagger\tpmodul9_2311104065/.vs/tpmodul9_2311104065/FileContentIndex/08a7fda9-cla8-459e-b432-4a3283f63cb6.vsidx
delete mode 100644 09_API_Design_dan_Construction_Using_Swagger\tpmodul9_2311104065/.vs/tpmodul9_2311104065/FileContentIndex/1f703fb6-fe09-46eb-b858-b535c0edaa2c.vsidx
delete mode 100644 09_API_Design_dan_Construction_Using_Swagger\tpmodul9_2311104065/.vs/tpmodul9_2311104065/FileContentIndex/306abfa1-44e0-49bb-9ca9-3776f08dd764.vsidx
delete mode 100644 09_API_Design_dan_Construction_Using_Swagger\tpmodul9_2311104065/.vs/tpmodul9_2311104065/FileContentIndex/3ed231b5-e0a2-469d-8abb-f561b215c9d0.vsidx
delete mode 100644 09_API_Design_dan_Construction_Using_Swagger\tpmodul9_2311104065/.vs/tpmodul9_2311104065/FileContentIndex/86ad0b6e-84ab-4c1f-a479-4ed48d400a50.vsidx
create mode 100644 09_API_Design_dan_Construction_Using_Swagger\tpmodul9_2311104065/.vs/tpmodul9_2311104065/FileContentIndex/f02216fa-592e-47c4-90c5-e94917138660.vsidx

C:\Users\user\source\repos\KPL_Rengganis-Tantri-Pramudita_2311104065_SISE-07-02\09_API_Design_dan_Construction_Using_Swagger\tpmodul9_2311104065>git push -u origin main
Enumerating objects: 40, done.
Counting objects: 100% (40/40), done.
Delta compression using up to 12 threads
```

- Mahasiswa.cs

```
namespace tpmodul9_2311104065.Models
```

```
{
    public class Mahasiswa
    {

        public string Nama { get; set; }
        public string Nim { get; set; }
    }
}
```

```
}  
}
```

Penjelasan:

Kode di atas mendefinisikan class Mahasiswa yang berada di dalam namespace tpmodul9_2311104065.Models. Class ini memiliki dua properti publik, yaitu Nama dan Nim, yang masing-masing bertipe string. Class ini digunakan sebagai model data untuk merepresentasikan objek mahasiswa dalam API yang ada di MahasiswaController. Properti get; set; memungkinkan data ini bisa dibaca dan dimodifikasi, baik saat menerima input dari pengguna (melalui POST) maupun saat mengembalikannya sebagai respons (melalui GET).

- MahasiswaController.cs

```
using Microsoft.AspNetCore.Mvc;  
using System.Collections.Generic;  
using tpmodul9_2311104065.Models;  
  
namespace tpmod9_2311104042.Controllers  
{  
    [ApiController]  
    [Route("api/[controller]")]  
    public class MahasiswaController : ControllerBase  
    {  
  
        private static List<Mahasiswa> daftarMahasiswa = new List<Mahasiswa>  
        {  
            new Mahasiswa { Nama = "rengganis", Nim = "2311104065" },  
            new Mahasiswa { Nama = "suga", Nim = "2311104001" },  
            new Mahasiswa { Nama = "jimin", Nim = "2311104076" }  
        };  
  
        [HttpGet]  
        public ActionResult<List<Mahasiswa>> Get()  
        {  
            return daftarMahasiswa;  
        }  
    }  
}
```

```

    }

    [HttpGet("{index}")]
    public ActionResult<Mahasiswa> Get(int index)
    {
        if (index < 0 || index >= daftarMahasiswa.Count)
            return NotFound();
        return daftarMahasiswa[index];
    }

    [HttpPost]
    public ActionResult<List<Mahasiswa>> Post([FromBody] Mahasiswa mhs)
    {
        daftarMahasiswa.Add(mhs);
        return daftarMahasiswa;
    }

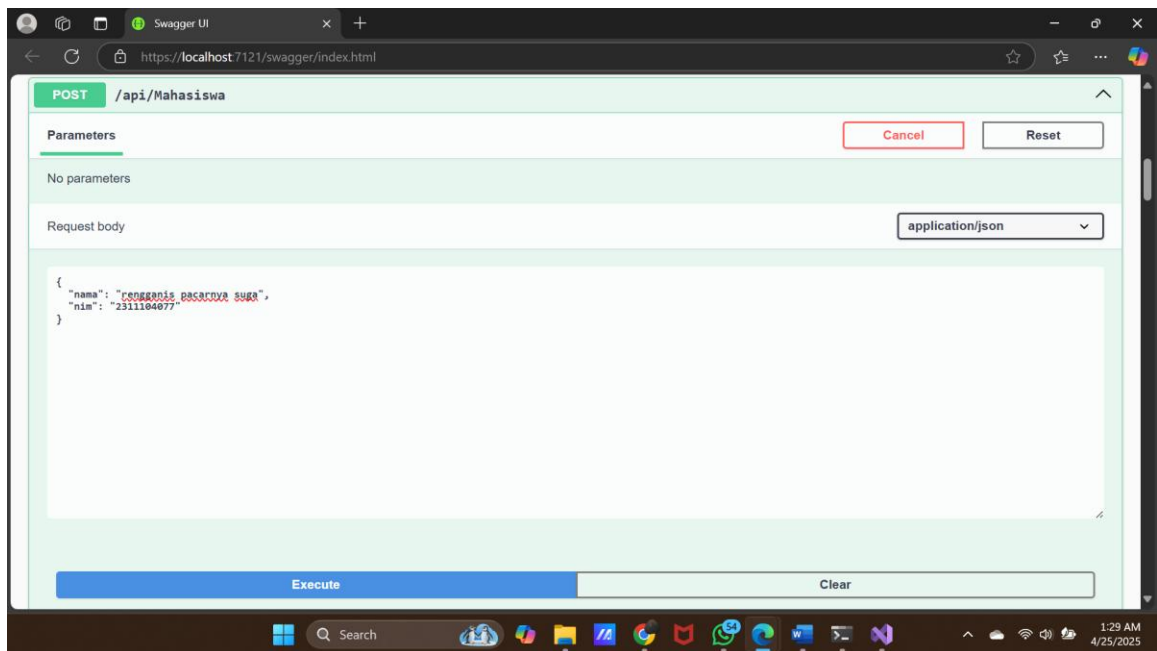
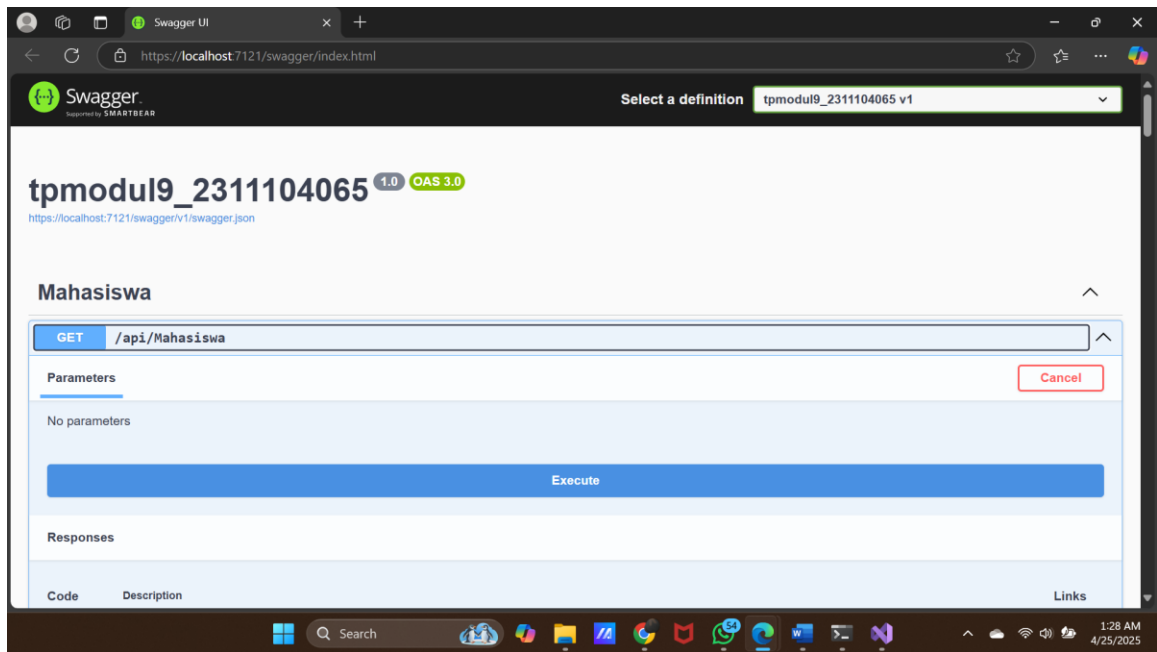
    [HttpDelete("{index}")]
    public ActionResult<List<Mahasiswa>> Delete(int index)
    {
        if (index < 0 || index >= daftarMahasiswa.Count)
            return NotFound();
        daftarMahasiswa.RemoveAt(index);
        return daftarMahasiswa;
    }
}

```

Penjelasan:

Kode di atas adalah implementasi API sederhana menggunakan ASP.NET Core yang menangani data mahasiswa. API ini berada dalam MahasiswaController dan menggunakan route api/mahasiswa. Data mahasiswa disimpan dalam list statis daftarMahasiswa. API ini memiliki empat endpoint: GET tanpa parameter untuk mengambil seluruh data mahasiswa, GET dengan parameter index untuk mengambil data mahasiswa tertentu, POST untuk menambahkan mahasiswa baru ke daftar, dan DELETE untuk menghapus mahasiswa berdasarkan index. Validasi sederhana juga dilakukan untuk mencegah akses atau penghapusan data di luar indeks yang tersedia.

- Hasil running



Swagger UI

https://localhost:7121/swagger/index.html

Responses

Curl

```
curl -X 'POST' \
  'https://localhost:7121/api/Mahasiswa' \
  -H 'accept: text/plain' \
  -H 'content-type: application/json' \
  -d '{
    "nama": "rengganis pacarnya suga",
    "nim": "2311104077"
  }'
```

Request URL

https://localhost:7121/api/Mahasiswa

Server response

Code	Details
200	Response body <pre>{ { "nama": "rengganis", "nim": "2311104065" }, { "nama": "suga", "nim": "2311104001" }, { "nama": "jimin", "nim": "2311104076" } }</pre>

1:30 AM
4/25/2025

Swagger UI

https://localhost:7121/swagger/index.html

```
},
{
  "nama": "jimin",
  "nim": "2311104076"
},
{
  "nama": "rengganis pacarnya suga",
  "nim": "2311104077"
}
]
```

Download

Response headers

```
content-type: application/json; charset=utf-8
date: Thu, 24 Apr 2025 18:26:59 GMT
server: Kestrel
```

Responses

Code	Description	Links
200	OK	No links

Media type

text/plain

Controls Accept header.

Example Value | Schema

```
{
  {
    "nama": "string",
    "nim": "string"
  }
}
```

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4/25/2025

Swagger UI

https://localhost7121/swagger/index.html

GET /api/Mahasiswa/{index}

Parameters

Cancel

Name	Description
index * required integer(int32) (path)	2

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'https://localhost:7121/api/Mahasiswa/2' \
  -H 'accept: text/plain'
```

Request URL

https://localhost:7121/api/Mahasiswa/2

Server response

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4/25/2025

Swagger UI

https://localhost7121/swagger/index.html

Server response

Code	Details
200	Response body <pre>{ "nama": "jimin", "nim": "2311104076"}</pre> Download Response headers <pre>content-type: application/json; charset=utf-8date: Thu, 24 Apr 2025 18:27:19 GMTserver: Kestrel</pre>

Responses

Code	Description	Links
200	OK	No links

Media type

text/plain

Controls Accept header

Example Value | Schema

```
{  "nama": "string",  "nim": "string"}
```

1:31 AM
4/25/2025

Swagger UI

https://localhost:7121/swagger/index.html

DELETE /api/Mahasiswa/{index}

Parameters

Cancel

Name	Description
index * required integer(int32) (path)	1

Execute Clear

Responses

Curl

```
curl -X 'DELETE' \
  'https://localhost:7121/api/Mahasiswa/1' \
  -H 'accept: text/plain'
```

Request URL

```
https://localhost:7121/api/Mahasiswa/1
```

Server response

Swagger UI

https://localhost:7121/swagger/index.html

-H 'accept: text/plain'

Request URL

```
https://localhost:7121/api/Mahasiswa/1
```

Server response

Code	Details
200	Response body <pre>{ { "nama": "rengganis", "nim": "2311104065" }, { "nama": "jimin", "nim": "2311104076" }, { "nama": "rengganis pacarnya suga", "nim": "2311104077" } }</pre> Download Response headers <pre>content-type: application/json; charset=utf-8 date: Thu, 24 Apr 2025 18:27:39 GMT server: Kestrel</pre>

Responses

Code	Description	Links
200	OK	No links

Swagger UI

https://localhost:7121/swagger/index.html

Responses

Code	Description	Links
200	OK	No links

Media type

text/plain

Controls Accept header

Example Value | Schema

```
{
  "nama": "string",
  "nim": "string"
}
```